# GPU Simulation of 2D Quantum Wavepackets

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# The Time-Dependent Schrödinger Equation (TDSE)

The Time-Dependent Schrödinger equation is a partial differential equation that describes the evolution of wave functions $\Psi$ over time.

### Time-Dependent Schrödinger equation

$$i\hbar\frac{\partial\Psi(x,y,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x,y)\right]\Psi(x,y,t). \tag{1}$$

Some example quantum systems in 2D include a free particle in a box, a finite potential barrier, and a potential barrier with double slit. These examples provides two test cases and an experimental case.

# Graphics Processing Units (GPU)

Graphics processing units are not just used for computer graphics and gaming. They also allow for running compute programs directly on the device, using a framework like Vulkan, Cuda or OpenCL.


Figure: Nvidia RTX 3070, a commonly available consumer GPU

# Wave Packet Generation

To solve the (TDSE), it is necessary to define the initial conditions.
In quantum mechanics, a localized particle can be prescribed by a linear superposition of plane waves.

## Wave Packet

$$\Psi(x, y, 0) = A \iint_{-\infty}^{\infty} P(p_x, p_y) \exp\left[\frac{i}{\hbar}\left(p_x(x - x_0) + p_y(y - y_0)\right)\right] dp_x\, dp_y$$

## Gaussian Momentum Distribution

$$P(p_x, p_y) = B \exp\left[-\frac{(p_x - p_{x_0})^2}{2\sigma_{p_x}^2} - \frac{(p_y - p_{y_0})^2}{2\sigma_{p_y}^2}\right]$$

# Wave Packet Generation Cont.

Finally, we assume maximal simultaneous localization, and use the classical energy-momentum relation to construct the initial condition

## 2D Gaussian Wave Packet

$$\Psi(x,y,0) = C \exp\left[ i\frac{\sqrt{2mE_0}}{\hbar} \left[(x-x_0)\cos\alpha + (y-y_0)\sin\alpha\right] \right.$$
$$\left. - \frac{1}{4\sigma_x^2}\left[(x-x_0)^2 + (y-y_0)^2\right] \right]$$

# Spatial Discretization

To approximate $\Psi(x, y)$ along its domain, we use a 2D Finite difference method. We take $\Delta x = \Delta y$. We use the Nine-point stencil Laplace operator $\nabla_9^2$ for its greater isotropy over the central difference kernel.

## Discreet Laplace Operator

$$\nabla_9^2 \psi_{i,j} = \frac{1}{6\Delta x^2} [\psi_{i-1,j+1} + 4\psi_{i,j+1} + \psi_{i+1,j+1} + 4\psi_{i-1,j} \\ -20\psi_{i,j} + 4\psi_{i+1,j} + \psi_{i-1,j-1} + 4\psi_{i,j-1} + \psi_{i+1,j-1}]$$

We apply this discretization to the 2D Schrödinger equation (1).

## Discreet TDSE

$$i\hbar \frac{\partial \psi_{i,j}}{\partial t} = \left[ -\frac{\hbar^2}{2m} \nabla_9^2 + V_{i,j} \right] \psi_{i,j}. \tag{2}$$

# Spatial Discretization Cont.

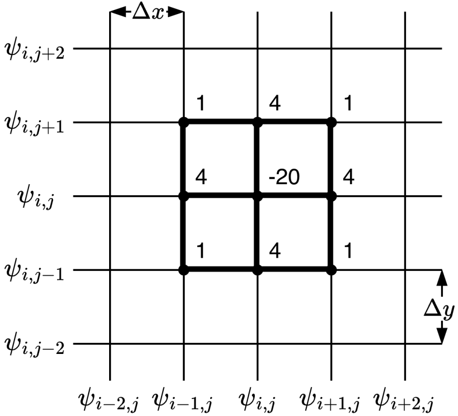



Figure: Graphical representation of the discretized domain, and $\nabla_9^2$ operator.

# Time Integration

Spatial discretization lets us treat the TDSE as a system of coupled ordinary differential equations. We use a second order, explicit Runge-Kutta method (RK2).

## 2nd Order Runge-Kutta

$$\begin{cases} \tilde{\psi}_{k+1} = \psi_k + (\Delta t) S(t_k, \psi_k) \\ \psi_{k+1} = \psi_k + \frac{\Delta t}{2} \left[ S(t_k, \psi_k) + S(t_{k+1}, \tilde{\psi}_{k+1}) \right] \end{cases} \tag{3}$$

We rewrite the TDSE once again

## TDSE

$$S := \frac{\partial \psi_{i,j}}{\partial t} = -\frac{i}{\hbar} \left[ -\frac{\hbar^2}{2m} \nabla_9^2 + V_{i,j} \right] \psi_{i,j}. \tag{4}$$

While RK2 does not inherently conserve certain quantities, we can choose a small enough time step such that these effects are negligible.

# Algorithmic Implementation

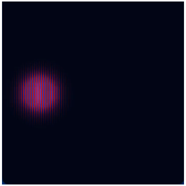
GPUs enable large-scale parallelism via SIMD (single instruction, multiple data). One GPU thread is dispatched per grid point. CPUs typically support roughly 10 hardware threads; GPUs support tens of thousands concurrently.

Parallel updates can cause race conditions when threads access the same memory. A 'ping-pong buffer' scheme ensures safe updates.

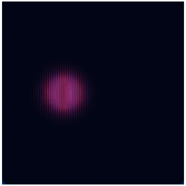
$$\psi_a \rightleftarrows \tilde{\psi} \rightleftarrows \psi_b$$

**Ping-Pong Buffers**

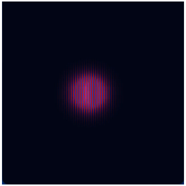
- Initial data in $\psi_a$.
- RK2 Stage 1: $\psi_a \to \tilde{\psi}$.
- RK2 Stage 2: $\tilde{\psi} \to \psi_b$.
- Next step reverses roles of $\psi_a$ and $\psi_b$.
- Alternation guarantees safe parallel writes at each time step.



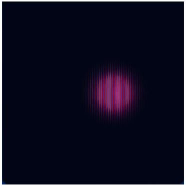
(a) $t = 0$ fs



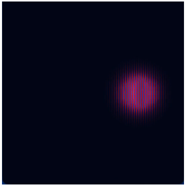
(b) $t = 2500$ fs



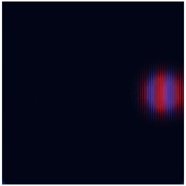
(c) $t = 5000$ fs



(d) $t = 7500$ fs

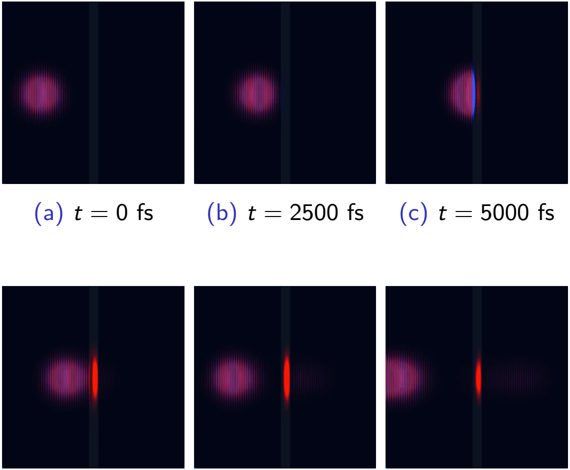


(e) $t = 10000$ fs



(f) $t = 12500$ fs

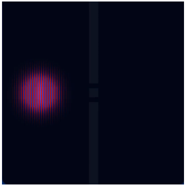
Figure: Simulation of a free electron wave packet. $E_0 = 0.01$ eV, $\alpha = 0$ rad, $\sigma_x = 50$ nm, $\Delta x = 1$ nm, $L = 1000$ nm, $\Delta t = 1$ fs. Real part of wave function, red is positive, blue is negative.



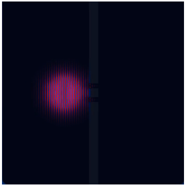
(a) $t = 0$ fs (b) $t = 2500$ fs (c) $t = 5000$ fs

(d) $t = 7500$ fs (e) $t = 10000$ fs (f) $t = 12500$ fs

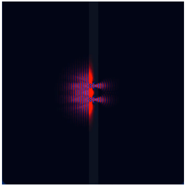
Figure: Simulation of an electron wave packet interacting with a potential barrier. $E_0 = 0.01$ eV, $\alpha = 0$ rad, $\sigma_x = 50$ nm, $\Delta x = 1$ nm, $L = 1000$ nm, $\Delta t = 1$ fs, $V_{barrier} = 0.01$ eV, $L_{barrier} = 50$ nm. Real part of wave function, red is positive, blue is negative.



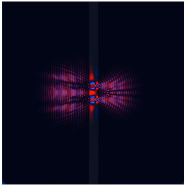
(a) $t = 0$ fs



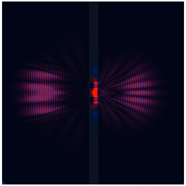
(b) $t = 2500$ fs



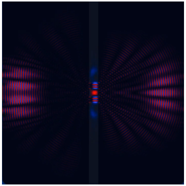
(c) $t = 5000$ fs



(d) $t = 7500$ fs

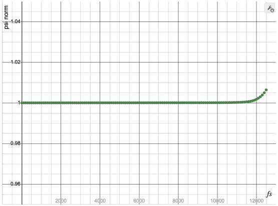


(e) $t = 10000$ fs

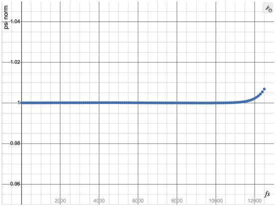


(f) $t = 12500$ fs

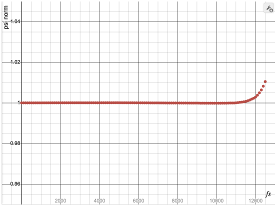
Figure: Simulation of an electron wave packet interacting with a double slit barrier. $E_0 = 0.01$ eV, $\alpha = 0$ rad, $\sigma_x = 50$ nm, $\Delta x = 1$ nm, $L = 1000$ nm, $\Delta t = 1$ fs, $V_{barrier} = 0.01$ eV, $L_{barrier} = 50$ nm, $d = 50$ nm, $a = 25$ nm. Real part of wave function, red is positive, blue is negative.




(a) Free particle (3)




(b) Potential barrier (4)




(c) Double slit barrier (5)

Figure: Wave function norms $|\Psi|^2$ over time (fs), corresponding to figures 3, 4, 5.

# References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